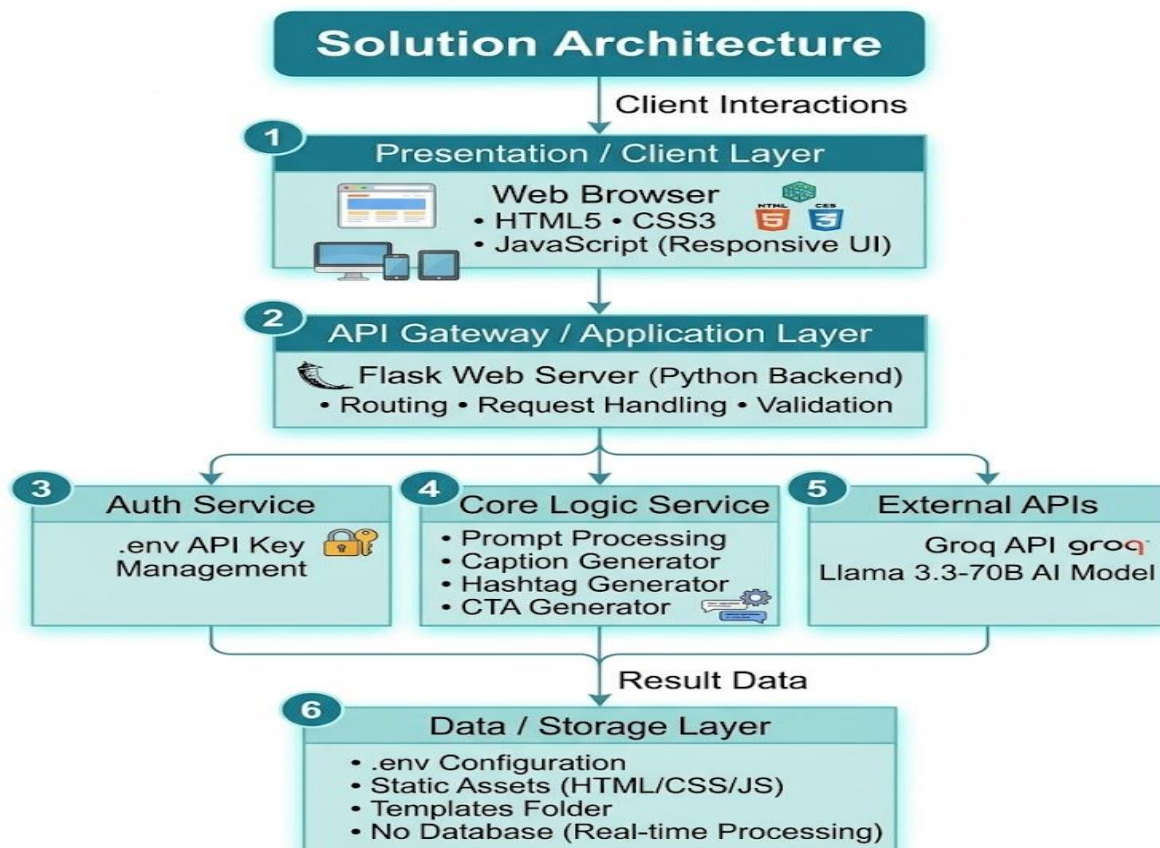


Solution Architecture

Date	02 July 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction Using Machine Learning
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>Presentation Layer</i>	<i>Provides the user interface for entering content, selecting platform and tone, and displaying AI-generated caption</i>	Provides the user interface for entering content, selecting platform and tone, and displaying AI-generated caption
<i>API Gateway / Application Layer</i>	<i>Receives user requests, validates inputs, routes requests to the AI service, and returns generated results.</i>	<i>Flask, Python</i>
Auth Service	<i>Securely stores and manages the Groq API key using environment variables.</i>	python-dotenv, .env
Core Logic Service	Generates prompts, communicates with Groq API, processes responses, and formats captions, hashtags, and CTAs.	<i>Python, Flask</i>
External API Integration	Uses the Groq LLM to generate AI-powered social media content.	<i>Groq API, Llama 3.3-70B Versatile</i>
Data / Storage Layer	Stores configuration files and application assets. No persistent database is required because all requests are processed in real time.	<i>.env, Static Files</i>